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Assignment 4 (SQL)

Load the provided dataset into your database

```sql
CREATE DATABASE TechCompanyDB;

USE TechCompanyDB;
```

- Open the provided links and download the dataset files

Items

Company

Employees

Departments

Questions

1. From the following tables, write a SQL query to calculate the average price of each company's products along with the company name. Return company_name and average_price.

```sql
SELECT    C.company_name,
          AVG(I.price) AS [Average Price]
FROM      company AS C
          INNER JOIN
          items AS I
          ON I.company_id = C.company_id
GROUP BY C.company_name;
```

2. Write a SQL query to find all companies where the average item price is greater than the overall average price of all products. Return company_name and average_price.

```sql
SELECT    C.company_name,
          AVG(I.price) AS [Average Price]
FROM      company AS C
          INNER JOIN
          items AS I
          ON I.company_id = C.company_id
GROUP BY C.company_name
HAVING    AVG(I.price) > (SELECT AVG(price)
                          FROM   items);
```

Questions

3. From the following tables, write a SQL query to find all items whose price is greater than the average price of all items. Return item_id, item_name, price, and company_id.

```
SELECT item_id,
       item_name,
       price,
       company_id
FROM   items
WHERE  price > (SELECT AVG(price)
                FROM   items);
```

4. From the following tables, write a SQL query to find all employees who work in departments whose department budget is greater than the average department budget. Return emp_id, emp_fname, emp_lname, and dept_code.

```
SELECT E.emp_id,
       E.emp_fname,
       E.emp_lname,
       D.dept_code
FROM   employees AS E
       INNER JOIN
       departments AS D
       ON D.dept_code = E.emp_dept
WHERE  D.dept_budget > (SELECT AVG(dept_budget)
                        FROM   departments);
```

Questions

5. Write a SQL query to find departments which have more employees than the average number of employees per department. Return dept_code and dept_name.

```sql
SELECT    D.dept_code,
          D.dept_name
FROM      departments AS D
          INNER JOIN
          employees AS E
          ON E.emp_dept = D.dept_code
GROUP BY D.dept_code, D.dept_name
HAVING    COUNT(E.emp_id) > (SELECT AVG(Total_Employee)
                             FROM   (SELECT   COUNT(emp_id) AS Total_Employee
                                     FROM     employees
                                     GROUP BY emp_dept) AS Emp_Count);
```

6. From the following tables, write a SQL query to find all items whose price is less than the average price of items for the same company. Return item_id, item_name, price, and company_id.

```sql
SELECT I1.item_id,
       I1.item_name,
       I1.price,
       I1.company_id
FROM   items AS I1
WHERE  I1.price < (SELECT AVG(I2.price) AS [Average Price]
                   FROM   items AS I2
                   WHERE  I1.company_id = I2.company_id);
```

Questions

7. Write a SQL query to find departments that have only one employee assigned to them. Return dept_code, dept_name, and the count of employees.

```sql
SELECT    D.dept_code,
          D.dept_name,
          COUNT(E.emp_id) AS Total_Employees
FROM      departments AS D
          INNER JOIN
          employees AS E
          ON E.emp_dept = D.dept_code
GROUP BY D.dept_code, D.dept_name
HAVING    COUNT(E.emp_id) = 1;
```

8. From the following tables, write a SQL query to find companies whose average item price is equal to the minimum average price among all companies. Return company_name and average_price.

```sql
SELECT    C.company_name,
          AVG(I.price) AS Average_Price
FROM      company AS C
          INNER JOIN
          items AS I
          ON I.company_id = c.company_id
GROUP BY C.company_name
HAVING    AVG(I.price) = (SELECT MIN(Average_Price)
                          FROM    (SELECT   AVG(price) AS Average_Price
                                   FROM     items
                                   GROUP BY company_id) AS AVG_Price);
```

9. Write a SQL query to find employees whose department budget is less than the average department budget. Return emp_id, emp_fname, emp_lname, and dept_code.

```sql
SELECT E.emp_id,
       E.emp_fname,
       E.emp_lname,
       D.dept_code
FROM   employees AS E
       INNER JOIN
       departments AS D
       ON D.dept_code = E.emp_dept
WHERE  D.dept_budget < (SELECT AVG(dept_budget)
                        FROM   departments);
```

Questions

1. Create a CTE that calculates the average item price, then display all items whose price is greater than the average.

```sql
WITH    AVG_items
AS      (SELECT AVG(price) AS Average_Price
         FROM   items)
SELECT *
FROM    items AS I CROSS JOIN AVG_items AS A
WHERE   I.price > A.Average_Price;
```

2. Create a CTE that calculates the total number of items per company, then display company name and item count.

```sql
WITH    Item_Count
AS      (SELECT   C.company_name,
                  COUNT(i.item_id) AS Total_items
         FROM     company AS C
                  INNER JOIN
                  items AS I
                  ON I.company_id = C.company_id
         GROUP BY C.company_name)
SELECT *
FROM    Item_Count;
```

Questions

3. Using a CTE, calculate the average price per company, then display companies where the average price is greater than 2000.

```sql
WITH    AVG_Price
AS      (SELECT    company_id,
                   AVG(price) AS AVG_P
         FROM      items
         GROUP BY company_id)
SELECT C.company_name,
       A.AVG_P
FROM   company AS C
       INNER JOIN
       AVG_Price AS A
       ON A.company_id = C.company_id
WHERE  A.AVG_P > 2000;
```

4. Create a CTE that joins employees_new and departments_new to display employee full name and department name.

```sql
WITH   NEW
AS     (SELECT CONCAT(E.emp_fname, ' ', E.emp_lname) AS Full_Name,
               D.dept_name
        FROM   employees AS E
               INNER JOIN
               departments AS D
               ON D.dept_code = E.emp_dept)
SELECT Full_Name,
       dept_name
FROM   NEW;
```

Questions

5. Using a CTE, find departments where the department budget is greater than the average budget.

```sql
WITH    AVG_Bud
AS      (SELECT AVG(dept_budget) AS AVG_Budget
         FROM   departments)
SELECT  D.dept_code,
        D.dept_name,
        D.dept_budget
FROM    departments AS D CROSS JOIN AVG_Bud AS A
WHERE   D.dept_budget > A.AVG_Budget;
```

6. Create a CTE that calculates the number of employees per department, then display departments that have more than one employee.

```sql
WITH    Count_Emp
AS      (SELECT    D.dept_name,
                   COUNT(E.emp_id) AS Total_Employees
         FROM      departments AS D
                   INNER JOIN
                   employees AS E
                   ON E.emp_dept = D.dept_code
         GROUP BY D.dept_name)
SELECT *
FROM    Count_Emp
WHERE   Total_Employees > 1;
```

7. Create a CTE that calculates the average item price across all companies, then display companies whose average item price is below the overall average.

```sql
WITH    AVG_item
AS      (SELECT    company_id,
                   AVG(price) AS Average_Price
         FROM      items
         GROUP BY company_id),
        All_AVG
AS      (SELECT AVG(price) AS OverAll_AVG
         FROM   items)
SELECT  C.company_name,
        A.Average_Price
FROM    company AS C
        INNER JOIN
        AVG_item AS A
        ON C.company_id = A.company_id CROSS JOIN All_AVG AS AG
WHERE   A.Average_Price < AG.OverAll_AVG;
```